

1 Aerobic respiration involves four main stages:

glycolysis, link reaction, Krebs cycle and oxidative phosphorylation.

(a) One product of all three stages shown in the table is ATP (adenosine triphosphate).

Complete the table to show one other product of each stage.

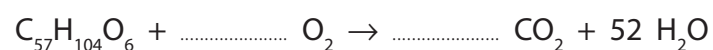
(3)

Stage	One other product
glycolysis	
Krebs cycle	
oxidative phosphorylation	

(b) An incomplete equation for the aerobic respiration of a substrate is shown below.

Complete the equation by inserting the numbers on the dotted lines to balance it.

(2)



(c) One molecule of glucose contains 2867.48 kJ of energy.

One molecule of glucose generates a maximum of 38 molecules of ATP in aerobic respiration.

A molecule of ATP contains 30.51 kJ of usable energy. Usable energy is available for chemical reactions in the cell.

(i) Which reaction releases energy from ATP?

(1)

- ☐ **A** the conversion of ADP to ATP
- ☐ **B** the phosphorylation of ADP
- ☐ **C** the hydrolysis of ATP
- ☐ **D** the removal of adenosine molecules from ATP

(ii) Calculate the maximum percentage of energy in a glucose molecule that can be converted to usable energy in ATP.

(2)

Answer%

(iii) Describe what happens to the energy that is not converted to usable energy in a muscle cell.

(2)

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(d) Describe how oxygen is involved in the production of ATP on the cristae in the mitochondria.

(2)

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(Total for Question 1 = 12 marks)
